```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split


df = pd.read_csv("/content/data.csv")
```

```
print(df.shape)
```

```
(944, 10)
```

```
print(df.isnull().sum())
```

```
footfall       0
tempMode       0
AQ             0
USS            0
CS             0
VOC            0
RP             0
IP             0
Temperature    0
fail           0
dtype: int64
```

```
X = df.drop("fail", axis=1)
y = df["fail"]

print(f"Shape of X: {X.shape}")
print(f"Shape of y: {y.shape}")
```

```
Shape of X: (944, 9)
Shape of y: (944,)
```

```
# Step 5: Scale numeric features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print("Shape:", X_scaled.shape)
```

```
Shape: (944, 9)
```

```
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42)
```

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
model = LogisticRegression()
model.fit(X_train, y_train)


y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.90      0.85      0.87       102
           1       0.84      0.89      0.86        87

    accuracy                           0.87       189
   macro avg       0.87      0.87      0.87       189
weighted avg       0.87      0.87      0.87       189
```